EAS-PQC: Analyzing Post Quantum Cryptography Algorithm Selection for IoT Device Classes

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***Abstract*—The migration of Internet of Things (IoT) systems to post-quantum cryptography (PQC) is essential to ensure long-term security against quantum enabled adversaries. However, the large computational, memory, and communication overheads of PQC algorithms pose challenges for resource-constrained IoT environments. In practice, IoT platforms span a wide range of capabilities, from battery-powered sensors to gateway devices, making a one-size-fits-all configuration impractical. This paper presents a new methodology for selecting suitable PQC schemes based on RFC 7228 device classes. We introduce the Efficiency-Adjusted Security for Post-Quantum Cryptography (EAS-PQC) metric that considers security strength, latency, energy consumption, memory footprint, communication overhead, and protocol reliability. Device-class-specific constraints and weighting profiles are incorporated to reflect the operational priorities of Class 0, Class 1, and Class 2 IoT devices. Our evaluation primarily uses performance benchmarks from the PQM4 framework, which are based on ARM Cortex-M4 microcontroller profiles, and performance results from several work on similar embedded platforms. Energy consumption is estimated using a constant active power model derived from Cortex-M4 power characteristics. We evaluate multiple PQC candidates, including ML-KEM and ML-DSA, and the results show a quantitative mapping between IoT device classes and PQC schemes. Specifically, Saber-LightSaber is identified as the most suitable option for Class 0 devices, while ML-DSA-87 combined with ML-KEM-768 achieves the highest overall scores for Class 1 and Class 2 devices. Our findings improve feasibility and efficiency by up to 30–40% compared to static baseline configurations. With these findings, we recommend EAS-PQC as a useful tool for the assessment of post-quantum security in industrial IoT systems.**

***Index Terms*—Post quantum cryptography, Efficiency-Adjusted Security (EAS), Internet of Things security, NIST PQC, security metrics.**

I. Introduction

The Internet of Things (IoT) is now part of many industrial systems, from field sensors and actuators to edge gateways and cloud-connected controllers. These devices are expected to run for years in environments with very limited memory, processing power, and energy. At the same time, the progress in quantum computing means that today's public-key schemes such as RSA and elliptic-curve cryptography will not be secure in the long term. The 'harvest now, decrypt later' threat makes this a current problem, not a distant one.

To address this, NIST has standardized a first set of post-quantum cryptography (PQC) algorithms: the module-lattice-based key encapsulation mechanism ML-KEM (Kyber), the module-lattice-based signature scheme ML-DSA (Dilithium), and the stateless hash-based signature scheme SLH-DSA (SPHINCS+) [1]–[3]. These schemes are designed to withstand quantum attacks, but they are significantly heavier than classical public-key algorithms in terms of key sizes, message sizes, computation time, and memory usage.

In this work we focus on a question: *given this metric and dataset, which NIST PQC schemes are best suited for Class 0, Class 1, and Class 2 IoT devices?* We adopt the RFC 7228 classification of constrained devices into Class 0, Class 1, and Class 2 [4], and we look at typical industrial scenarios such as battery-powered sensors, industrial control actuators, and gateway devices.

To do this, we introduce the Efficiency-Adjusted Security for Post-Quantum Cryptography (EAS-PQC) metric that evaluates each candidate configuration along six dimensions: security level, latency, energy consumption, memory footprint, communication overhead, and protocol reliability. Each metric is normalized and then combined with context-specific weights into a single score. The metric is driven by the benchmark data from PQM4 for ARM Cortex-M4 microcontrollers [5] and from several work on KEMs and signatures [6]–[10]. While our work employs a quantitative mathematical methodology for PQC selection, future extensions may incorporate AI-driven methods for intelligent weight optimization and device classification, to align the work more closely with AI-based cybersecurity approaches.

The contributions of this paper are as follows:

- We formalize an evaluation methodology for PQC in

IoT, named as EAS-PQC. The methodology combines six metrics (security, latency, energy, memory, overhead, reliability) into a single score with device-class-specific weights.
- We build a dataset based on PQM4 benchmarks and 10 other research work, covering ML-KEM, ML-DSA, Saber, and related schemes on Cortex-M4 and similar platforms.
- We derive a quantitative mapping from IoT device classes to PQC schemes. For Class 0 devices, Saber-LightSaber emerges as the only practical and high-scoring choice. For Class 1 and Class 2 devices, ML-DSA-87 combined with ML-KEM-768 provides the best trade-off between security and cost.
- We compare these choices against a static 'one-size-fits-all' baseline and show that selections based on our EAS-PQC can improve feasibility and overall score by around 30–40%, especially for the most constrained devices.

II. Related Work

Liu et al. [11] survey over fifty PQC contributions for resource-constrained devices, covering implementation challenges and performance trends, but do not map specific schemes to RFC 7228 device classes.

Fitzgibbon and Ottaviani [6] benchmark ML-KEM and ML-DSA on Raspberry Pi, measuring latency, energy, and memory for a Class 2 platform. Abbasi et al. [7] extend benchmarking to heterogeneous environments including servers and embedded boards. Lopez et al. [12] evaluate PQC on resource-constrained devices with detailed per-microcontroller metrics, and Hanna et al. [13] study PQC performance on consumer IoT devices with realistic protocol stacks.

Halak et al. [8] evaluate energy and computation costs of quantum-resistant algorithms for embedded devices, showing significant overheads compared to classical schemes. Patterson et al. [9] develop an energy-consumption framework for post-quantum key generation on embedded devices. Tasopoulos et al. [14] evaluate energy consumption of post-quantum TLS 1.3 on resource-constrained embedded devices.

Bürstinghaus-Steinbach et al. [15] integrate post-quantum KEM and signature schemes into embedded TLS and measure communication overhead on multiple platforms. Ehsan et al. [10] compare Kyber and NTRU in IoT scenarios, and Lucas et al. [16] propose a lightweight Ring-LWE hardware accelerator for PQC.

Multi-objective frameworks for security–performance trade-offs also exist. Liu et al. [11] review analytic hierarchy process and similar ranking methods. Temirbekova et al. [17] propose a multi-criteria evaluation for IoT cryptography, but their work is mostly qualitative and not tied to specific NIST PQC parameter sets.

These works highlight that no single metric suffices and that the bottleneck changes with device class. However, most studies keep metrics separate and are not linked to RFC 7228 device classes. To the best of our knowledge, no prior work (i) takes NIST-standardised PQC schemes, (ii) evaluates them using a multi-criteria metric based on real benchmark data, and (iii) produces a mapping from RFC 7228 classes to recommended PQC schemes. Our EAS-PQC metric fills this gap.

III. Background and Metric Definition

We follow the terminology of RFC 7228 [4]. It defines three main device classes based on available RAM and flash memory:

- **Class 0** (C0): Severely constrained devices with less than about 10 KB of RAM and 100 KB of flash.
- **Class 1** (C1): Moderately constrained devices with roughly 10–50 KB of RAM and up to a few hundred KB of flash.
- **Class 2** (C2): Less constrained devices with at least 50 KB of RAM and more generous flash and CPU resources.

In line with this classification, we consider three representative industrial scenarios: a Class 0 battery-powered field sensor, a Class 1 industrial control actuator, and a Class 2 IoT gateway that aggregates traffic from many devices and connects to the cloud. These scenarios match common industrial deployments [5]–[7].

A. PQC Schemes and Parameter Sets

NIST has standardised three main post-quantum schemes: the key-encapsulation mechanism ML-KEM, the signature scheme ML-DSA, and the hash-based signature scheme SLH-DSA [1]–[3]. For constrained devices, most practical work so far has focused on ML-KEM and ML-DSA, sometimes combined with alternative schemes that have smaller footprints, such as Saber and NTRU [5], [10].

In this paper, we concentrate on the schemes and parameter sets that have sufficient benchmark coverage and are relevant for IoT deployments:

- **ML-KEM**: parameter sets ML-KEM-512 and ML-KEM-768 (NIST security levels 1 and 3).
- **ML-DSA**: parameter sets ML-DSA-44, ML-DSA-65, and ML-DSA-87 (NIST security levels 2, 3, and 5).
- **Saber**: the LightSaber parameter set as a lightweight KEM candidate for very constrained devices.
- **Kyber-512**: an implementation equivalent to ML-KEM-512 in the PQM4 suite, used in some embedded benchmarks [5].

We consider both individual schemes and combined protocol configurations that pair a KEM with a signature scheme, for example ML-KEM-768 with ML-DSA-65, or LightSaber with a basic authentication mechanism. Performance and resource metrics are taken from PQM4 and complementary PQC studies on embedded systems where needed [5]–[9].

B. Our EAS-PQC Metric

The goal of the Efficiency-Adjusted Security for Post-Quantum Cryptography (EAS-PQC) metric is to combine several relevant metrics into a single score that can be used

TABLE I
NOTATION USED IN THE EAS-PQC METRIC

Symbol	Meaning
c	Candidate PQC configuration (KEM + signature)
S	Security level (bits)
L	Handshake latency (ms)
E	Energy per handshake (mJ)
M	Peak RAM usage (KB)
P	Communication overhead (bytes)
R	Handshake reliability (probability)
X_{norm}	Normalised value of metric $X \in \{S, L, E, M, P, R\}$
w_X	Weight assigned to metric X
$\mathrm{Score}(c)$	EAS-PQC score for candidate c

to rank candidate PQC configurations for a given device class and scenario.

For each candidate configuration c, EAS-PQC defines six sub-metrics:

- S: security level (in bits, mapped from NIST levels).
- L: latency of a complete cryptographic handshake (ms).
- E: energy consumption for the handshake (mJ).
- M: peak RAM usage during the handshake (KB).
- P: communication overhead, measured as the total size of keys, ciphertexts, and signatures (bytes).
- R: protocol reliability, expressed as the probability that the handshake completes successfully under the assumed network conditions.

Each raw metric is normalised to a value between $[0, 1]$. The final EAS-PQC score for candidate c is a weighted sum of the normalised sub-scores:

$$\begin{aligned}\mathrm{Score}(c) = w_S S_{\mathrm{norm}}(c) + w_L L_{\mathrm{norm}}(c) + w_E E_{\mathrm{norm}}(c) \\ + w_M M_{\mathrm{norm}}(c) + w_P P_{\mathrm{norm}}(c) + w_R R_{\mathrm{norm}}(c).\end{aligned} \tag{1}$$

$$w_S + w_L + w_E + w_M + w_P + w_R = 1. \tag{2}$$

The weights $(w_S, \ldots, w_R)$ reflect the priorities of the device class and application, and are chosen differently for Class 0, Class 1, and Class 2.

Table I summarises the notation used in the rest of the paper.

EAS-PQC uses a min–max normalisation scheme so that all sub-metrics lie in $[0, 1]$ and higher values are better. We treat security and reliability as *benefit* metrics (higher raw values are desirable), and latency, energy, memory, and overhead as *cost* metrics (lower raw values are desirable). For benefit criteria, where higher raw values are desirable (security and reliability), the normalised score is

$$X_{\mathrm{norm}} = \frac{X_{\mathrm{raw}} - X_{\mathrm{min}}}{X_{\mathrm{max}} - X_{\mathrm{min}}}, \tag{3}$$

for $X \in \{S, R\}$. For cost criteria, where lower raw values are desirable (latency, energy, memory, overhead), the normalised score is

$$X_{\mathrm{norm}} = 1 - \frac{X_{\mathrm{raw}} - X_{\mathrm{min}}}{X_{\mathrm{max}} - X_{\mathrm{min}}}, \tag{4}$$

for $X \in \{L, E, M, P\}$.

TABLE II
STANDARD WEIGHTING PROFILES PER DEVICE CLASS

Metric	Class 0 (sensor)	Class 1 (actuator)	Class 2 (gateway)
w_S (Sec.)	0.15	0.25	0.30
w_L (Lat.)	0.25	0.30	0.20
w_E (Energy)	0.25	0.15	0.10
w_M (Mem.)	0.15	0.10	0.10
w_P (Over.)	0.10	0.10	0.05
w_R (Rel.)	0.10	0.10	0.25
Total	1.00	1.00	1.00

The normalisation ranges $(X_{\mathrm{min}}, X_{\mathrm{max}})$ are computed from the set of candidates under consideration for each scenario.

In EAS-PQC, reliability is based on packet loss and the number of handshake messages. If p is the packet loss probability on the link used for the handshake, and m is the number of messages exchanged, the reliability R is

$$R = (1 - p)^m. \tag{5}$$

This assumes independent loss events and that any lost handshake message causes the protocol to fail. Four-message KEM-based handshakes therefore have lower reliability than two-message signature-based handshakes on a lossy link, for the same p. This effect is particularly relevant for Class 0 devices deployed over low-power wide-area networks.

C. Device-Class-Specific Weighting Profiles

To capture the different priorities of each device class, we use three standard weighting profiles based on RFC 7228 classes and common industrial scenarios. Table II shows the weights used for the experiments in this paper.

For Class 0, energy and latency receive high weight, reflecting battery constraints. For Class 1, security and latency dominate. For Class 2, security and reliability are prioritized, since gateways act as security boundaries. These profiles provide a consistent baseline for comparison but can be adjusted for specific deployments.

IV. EXPERIMENTAL SETUP

A. Data Sources and Hardware Model

Our experiments use benchmark data from the PQM4 project [5], which provides cycle counts, code size, and stack usage for NIST PQC candidates on ARM Cortex-M4. We assume a Cortex-M4 at 48 MHz. Cycle counts are converted to execution times ($t = C/f$) and energy is estimated using a constant active power model ($E = P_{\mathrm{active}} \cdot t$) following [6]–[9]. Where PQM4 lacks metrics, we supplement with data from [6], [7], [10]. Communication overhead is computed from NIST-documented key, ciphertext, and signature sizes [1], [2].

B. Candidate Algorithms and Configurations

The candidate set covers NIST-standardised schemes and lightweight alternatives:

- ML-KEM-512 and ML-KEM-768 (key encapsulation).
- ML-DSA-44, ML-DSA-65, and ML-DSA-87 (digital signatures).

TABLE III
DEVICE-CLASS CONSTRAINTS USED IN THE EXPERIMENTS

Constraint	Class 0 (sensor)	Class 1 (actuator)	Class 2 (gateway)
RAM limit [KB]	8–10	32–64	≥ 128
Energy per handshake	Tight	Moderate	Loose
Network type	LPWAN	Ethernet/fieldbus	LAN/WAN
Packet loss p	5–8%	1–2%	$< 1\%$
Max packet size [bytes]	small	medium	large

- LightSaber as a lightweight KEM candidate.
- Kyber-512 as implemented in PQM4, treated as equivalent to ML-KEM-512 where appropriate [5].

From these building blocks we form combined protocol configurations that represent a full key establishment and authentication flow. For example, a configuration may pair ML-KEM-768 with ML-DSA-65, or LightSaber with a signature scheme or a pre-shared-key based authenticator. Each configuration c is treated as a single candidate in the EAS-PQC evaluation and is assigned aggregate metrics (latency, energy, memory, overhead, reliability) for a complete handshake.

For comparison, we also define a static baseline configuration that is the same across device classes. In our experiments, this baseline is a typical mid-level setting based on ML-KEM-768 and ML-DSA-65, to reflect a straightforward application of NIST recommendations without class-specific tuning. Table III summarises the constraints used in the experiments.

For Class 0, we assume a battery-powered sensor node with 8–10 KB of RAM and a low-power wide-area network such as LoRaWAN, with a packet loss rate around 5–8% and strict limits on packet size and duty cycle. For Class 1, we assume an industrial control actuator with around 32–64 KB of RAM. For Class 2, we assume a gateway-class device with at least 128 KB of RAM and a reliable LAN or WAN connection with low packet loss.

These constraints are used in two ways. First, we apply a feasibility check to remove any candidate configuration whose memory footprint exceeds the available RAM, or whose communication overhead clearly violates the network limitations. Second, the network parameters (in particular the packet loss probability p and the number of handshake messages m) are used to compute the reliability metric $R = (1 - p)^m$ for each configuration.

C. EAS-PQC Computation Methodology

For each device class and scenario, the EAS-PQC score is computed in four steps:

1) **Constraint filtering:** We start from the full candidate set and discard any configuration that exceeds the RAM limit of the class or requires packet sizes that are not compatible with the assumed network. This step ensures that we only score configurations that are realistically deployable on the target devices.
2) **Metric aggregation:** For each remaining candidate, we compute the raw metrics S, L, E, M, P, and R as described in Section III. Security level is taken from the NIST parameters, while latency, energy, and memory are derived from benchmark data. Communication overhead and reliability are computed from key, ciphertext, and signature sizes and from the assumed network parameters.
3) **Normalisation:** For each metric we calculate $X_{\min}$ and $X_{\max}$ across the feasible candidates for the given scenario, and apply min–max normalisation to obtain X_{norm} in $[0, 1]$, with higher values representing better outcomes.
4) **Scoring and ranking:** We select the appropriate weighting profile for the device class (Table II) and compute the EAS-PQC score using (1). Candidates are then ranked by their score, and we record the top-performing configuration for each class, along with its feasibility status and individual metric values.

This procedure is repeated for all three device classes.

TABLE IV
TOP PQC CANDIDATES FOR CLASS 0 (ENERGY-FOCUSED)

Candidate	Sec. [bit]	Lat. [ms]	Energy [mJ]	Mem. [KB]	Over. [B]	Score
Saber-LightSaber	128	4.20	1.47	6.2	1472	0.875
Kyber-512	128	6.50	2.28	6.8	1568	0.853
ML-KEM-512	128	8.13	2.85	7.2	1568	0.838
ML-DSA-44	128	12.06	4.22	8.5	2420	0.807
ML-KEM-768	192	12.22	4.28	9.8	2272	0.804

V. RESULTS AND DISCUSSION

This section presents the EAS-PQC scores for the three device classes and discusses which PQC schemes emerge as suitable choices. We first describe the per-class rankings, then compare them with a static baseline configuration.

A. Class 0: Energy-Constrained Sensors

The Class 0 scenario models a battery-powered environmental sensor with 8–10 KB of RAM, strict energy limits, and a low-power wide-area network (LPWAN) link. After applying the constraints (RAM ≤ 10 KB, energy ≤ 100 mJ, overhead ≤ 5000 bytes), 11 out of 17 candidates remained feasible, giving a feasibility rate of 65%. The EAS-PQC weighting in this scenario favours energy and memory, reflecting the sensor's limited battery and RAM.

Table IV summarises the top five candidates for Class 0. Saber-LightSaber achieves the highest EAS-PQC score of 0.875, followed by Kyber-512 and ML-KEM-512 with scores of 0.853 and 0.838 respectively. All top candidates operate at 128-bit security, which is the only security level that fits within the memory constraints of this class.

Saber-LightSaber is selected as the most suitable scheme for Class 0 sensors because it combines very low energy consumption and small memory footprint with acceptable latency and overhead.

B. Class 1: Industrial Control Actuators

The Class 1 scenario represents an industrial Ethernet actuator with moderate resource constraints. The evaluation assumes up to 50 KB RAM, a maximum energy budget of 300 mJ per handshake, and a communication overhead limit of 8000 bytes.

TABLE V
TOP PQC CANDIDATES FOR CLASS 1 (LATENCY-FOCUSED)

Candidate	Sec. [bit]	Lat. [ms]	Energy [mJ]	Mem. [KB]	Over. [B]	Score
ML-DSA-87	256	28.70	10.05	15.4	4595	0.898
ML-KEM-1024	256	19.92	6.97	13.1	3136	0.807
ML-DSA-65	192	18.50	6.48	11.2	3293	0.789
ML-KEM-768	192	12.22	4.28	9.8	2272	0.687
ML-DSA-44	128	12.06	4.22	8.5	2420	0.665

TABLE VI
TOP PQC CANDIDATES FOR CLASS 2 (SECURITY-FOCUSED)

Candidate	Sec. [bit]	Lat. [ms]	Energy [mJ]	Mem. [KB]	Over. [B]	Score
ML-DSA-87	256	28.70	10.05	15.4	4595	0.914
ML-KEM-1024	256	19.92	6.97	13.1	3136	0.863
ML-DSA-65	192	18.50	6.48	11.2	3293	0.826
ML-KEM-768	192	12.22	4.28	9.8	2272	0.766
ML-KEM-1024 + ML-DSA-87 (cfg)	256	48.62	17.02	28.5	7731	0.729

The weighting profile gives equal priority to security and latency, reflecting the need for both cryptographic protection and timely control. It is found that 15 of 17 candidates remain feasible (88% feasibility). The top five candidates and their scores are shown in Table V. ML-DSA-87 achieves the highest EAS-PQC score of 0.898. It offers 256-bit security with good latency and high reliability. ML-KEM-1024 and ML-DSA-65 follow with scores of 0.807 and 0.789.

The ranking indicates that for Class 1 actuators, higher security levels are feasible and attractive when latency remains within the acceptable range. ML-DSA-87 is a good fit because the Class 1 memory and energy budgets can accommodate its larger parameters, and the weighting profile rewards its stronger security and high handshake reliability.

C. Class 2: IoT Gateways

The Class 2 scenario models an IoT gateway with more resources (up to 256 KB RAM). Here the weighting profile puts more weight on security and reliability, since gateways sit at the boundary between constrained networks and backend systems. With these constraints, all 17 candidates are feasible (100% feasibility). Table VI lists the top five candidates. ML-DSA-87 again achieves the highest EAS-PQC score, now 0.914. ML-KEM-1024 and ML-DSA-65 follow with scores of 0.863 and 0.826.

In this class, security is not limited by memory or energy, so 256-bit schemes dominate the top of the ranking. ML-DSA-87 benefits from both its higher security level and its strong reliability, while its resource costs remain acceptable for gateway hardware.

Across the three scenarios, the EAS-PQC framework consistently selects one best-scoring algorithm per device class. Table VII summarises the recommendations for signature schemes, while Table VIII presents the KEM+signature configurations proposed for practical deployment. The difference shows that signature-only evaluation prioritizes higher security levels (ML-DSA-87 for Classes 1 and 2), whereas KEM+signature configurations balance both KEM and signature overheads, leading to the combinations of ML-DSA-44 for Class 1 and ML-DSA-65 for Class 2, i.e., feasibility within device constraints is maintained.

TABLE VII
SUMMARY OF EAS-PQC ALGORITHM RECOMMENDATIONS BY IOT DEVICE CLASS

IoT class	Representative role	Recommended algorithm	NIST level	Score
Class 0 (C0)	Battery-powered sensor	Saber-LightSaber (KEM)	Level 1 (128-bit)	0.875
Class 1 (C1)	Industrial actuator	ML-DSA-87 (signature)	Level 5 (256-bit)	0.898
Class 2 (C2)	IoT gateway	ML-DSA-87 (signature)	Level 5 (256-bit)	0.914

TABLE VIII
RECOMMENDED KEM+SIGNATURE CONFIGURATIONS BY IOT DEVICE CLASS

IoT class	Recommended configuration	NIST level	Score
Class 0 (C0)	ML-KEM-512 only	Level 1 (128-bit)	0.78
Class 1 (C1)	ML-KEM-512 + ML-DSA-44	Level 1 (128-bit)	0.82
Class 2 (C2)	ML-KEM-768 + ML-DSA-65	Level 3 (192-bit)	0.89

D. Comparison with a Static Baseline

We compare our class based selections with a static baseline (ML-KEM-768 + ML-DSA-65, 192-bit security) that does not vary across device classes. Table IX summarises the results. For Class 0, the baseline is infeasible because its memory footprint exceeds the sensor's RAM limit; the transition from infeasible to feasible (Saber-LightSaber, score 0.875) shows an improvement that makes deployment feasible, where none was possible. For Class 1 and Class 2, the baseline scores 0.691 and 0.656 respectively, while ML-DSA-87 reaches 0.898 (+30%) and 0.914 (+39%).

TABLE IX
EAS-PQC RECOMMENDATIONS COMPARED TO A STATIC BASELINE

Scenario	Baseline config.	Baseline score	EAS-PQC rec.	Improv.
Class 0	ML-KEM-768 + ML-DSA-65	Infeasible	Saber-LightSaber	N/A
Class 1	ML-KEM-768 + ML-DSA-65	0.691	ML-DSA-87	+30.0%
Class 2	ML-KEM-768 + ML-DSA-65	0.656	ML-DSA-87	+39.3%

VI. CONCLUSION

This paper addressed the question of which PQC schemes suit which RFC 7228 IoT device classes. Using the EAS-PQC metric across six dimensions (security, latency, energy, memory, overhead, reliability) and benchmark data from PQM4 and related studies, we evaluated candidates for three industrial scenarios. Saber-LightSaber achieves the highest score for Class 0 due to its low energy and memory footprint. For Class 1 and Class 2, ML-DSA-87 performs best (scores 0.898 and 0.914), combining 256-bit security with acceptable latency. Compared with a static ML-KEM-768 + ML-DSA-65 baseline, the class-aware recommendations yield 30% and 39% score improvements for Class 1 and Class 2, while enabling Class 0 deployment where the baseline is infeasible.

Limitations and Future Work: Our evaluation based on ARM Cortex-M4 benchmarks (PQM4), which may not translate directly to RISC-V or 8/16-bit AVR architectures common in Class 0 devices. The energy model uses a simplified constant active power assumption ($E = P_{\text{active}} \cdot t$) that may miss the power spikes of specific operations or hardware acceleration. The weighting profiles (Table II) are grounded in industrial scenarios but remain baseline assumptions; a sensitivity analysis examining ranking stability under weight perturbation might be needed to strengthen confidence in the results. Our candidate set also focuses primarily on NIST-standardized schemes with limited exploration of other lightweight cryptographic primitives beyond Saber-LightSaber. Future work will expand platform coverage to RISC-V and AVR, refine energy models with profiles of several specific operations, conduct sensitivity analysis across weight perturbation ranges, and incorporate machine learning for weight optimization.

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